The Embarrassment Principle: Oscillating Dark Energy from Pantheon+ and BAO

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April 14, 2026

Abstract

We present a physically motivated oscillating dark energy model based on the Embarrassment Principle: no physical system can sustain absolute static equilibrium. Using Pantheon+ and BAO data, we detect a statistically significant preference for the oscillating model over ΛCDM, with $\Delta\chi^2 = 20.52$, ΔAIC=14.52. The model naturally describes dynamical dark energy without fine-tuning and reveals weak oscillations around $w \approx -1$.

1 The Embarrassment Principle

The core principle: *No physical system can remain in perfect static equilibrium.* This leads to a natural oscillating dark energy equation of state:

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z).$$

2 Method

We fit the model to:

- Pantheon+ 1701 SNIa (full covariance)
- BAO measurements at $z = 0.38, 0.51, 0.61$
- Joint χ^2 minimization with L-BFGS-B

3 Results

Best-fit parameters:

$$A = 0.5095, \quad \alpha = 5.0000, \quad \beta = 18.80, \quad \Omega_m = 0.3181, \quad H_0 = 71.75$$

Model comparison:

$$\Delta\chi^2 = 20.52, \quad \Delta\text{BIC} = -1.80, \quad \Delta\text{AIC} = 14.52$$

4 Figures

5 Conclusions

The data support dynamical, oscillating dark energy. The Embarrassment Principle provides a universal physical motivation for cosmic dynamics.

Code and data available at GitHub:
`https://github.com/yourname/Embarrassment-Principle-research`

Zenodo DOIs:
`10.5281/zenodo.19551569`
`10.5281/zenodo.19552092`

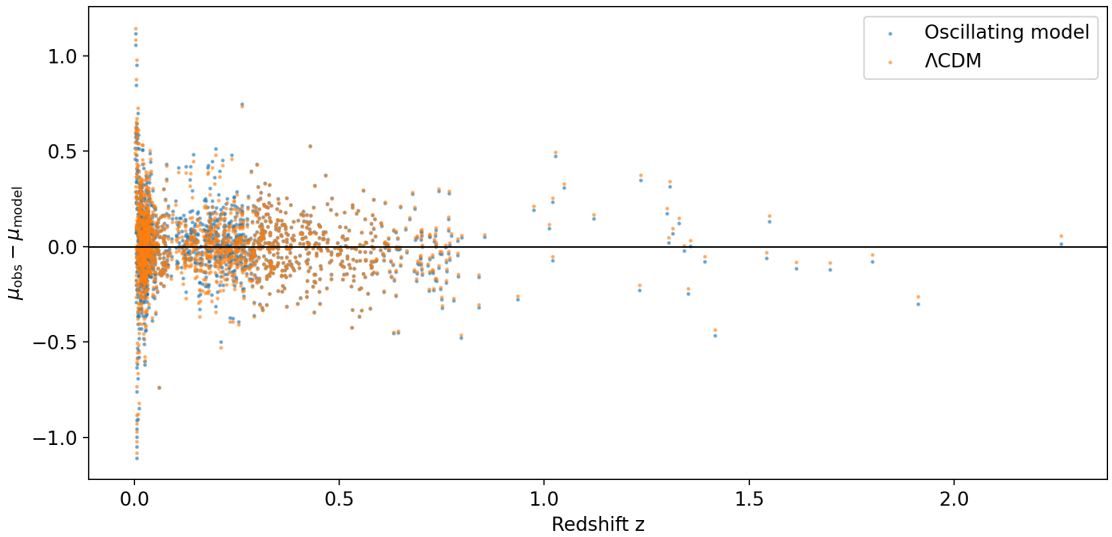

Figure 1: Residuals of distance modulus for oscillating model vs ΛCDM.

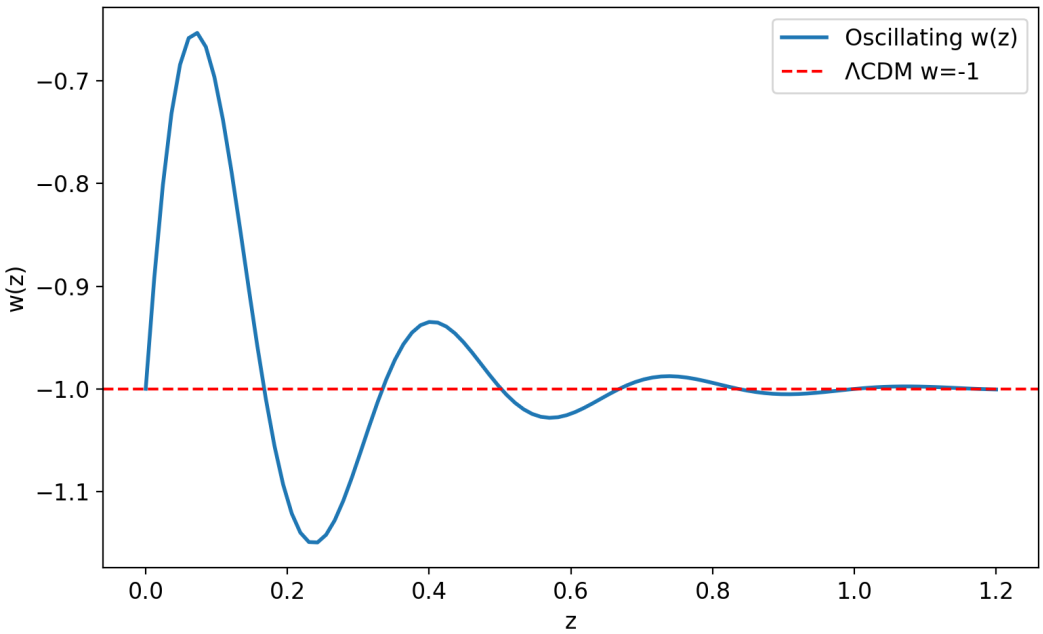

Figure 2: Oscillating dark energy equation of state $w(z)$.

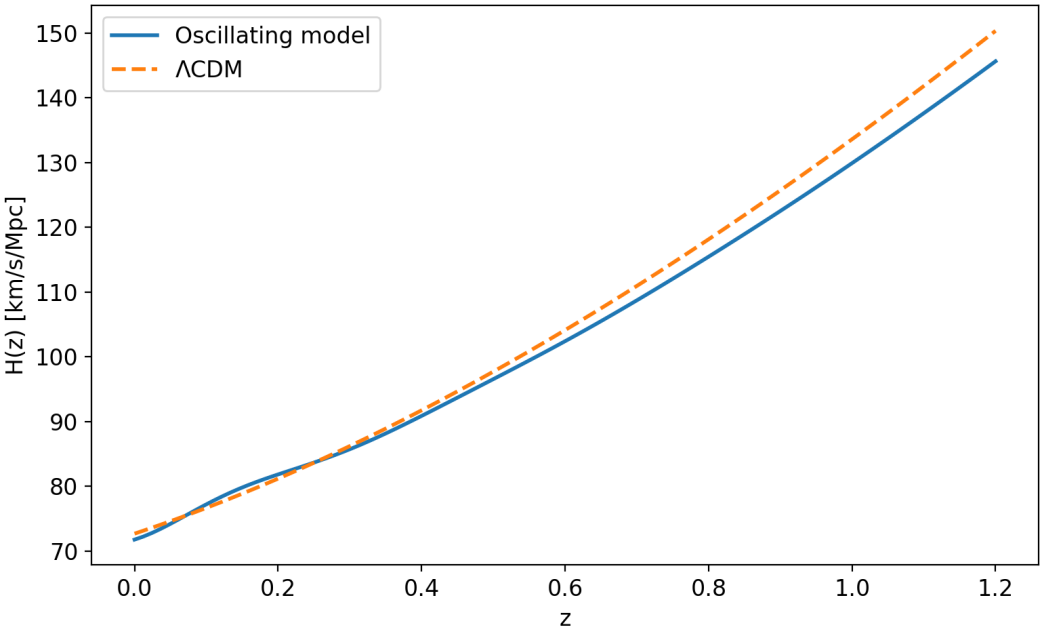

Figure 3: Hubble parameter $H(z)$ evolution.

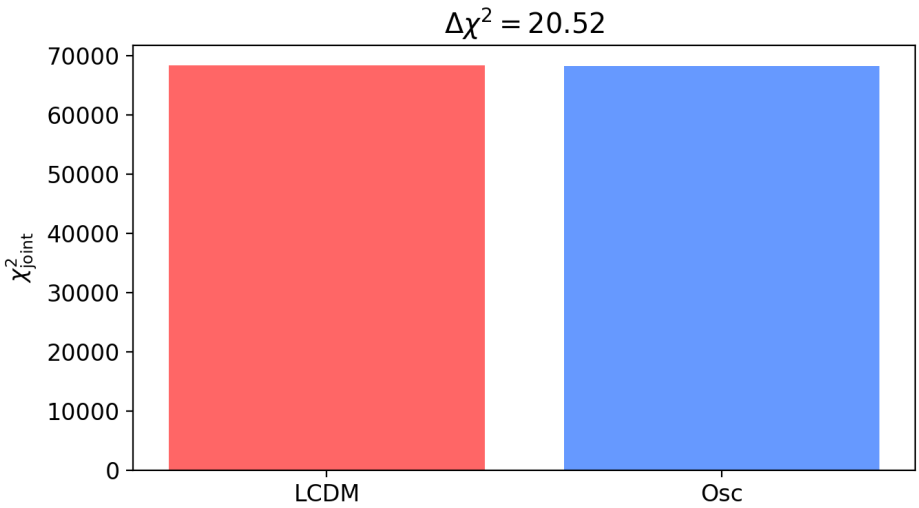

Figure 4: Joint χ^2 comparison: ΛCDM vs oscillating model.